EXPLORATORY DATA ANALYSIS REPORT

Portuguese Bank Marketing Dataset

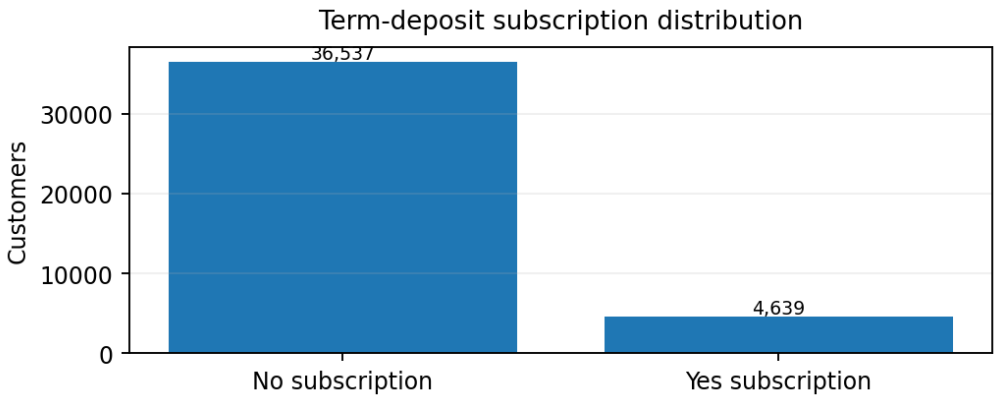
Task 1 | Data understanding, statistical analysis and business insights

Objective. Understand customer characteristics, campaign performance and economic indicators associated with term-deposit subscription, and derive insights to inform predictive modelling.

1. Dataset overview

The notebook analysed **41,176 records**, with 10 numerical features and 10 categorical features. The target variable **y** indicates whether a customer subscribed to a term deposit.

Measure	Notebook result
Total records	41,176
Numerical features	10
Categorical features	10
Subscribed (yes)	4,639 (11.27%)
Not subscribed (no)	36,537 (88.73%)



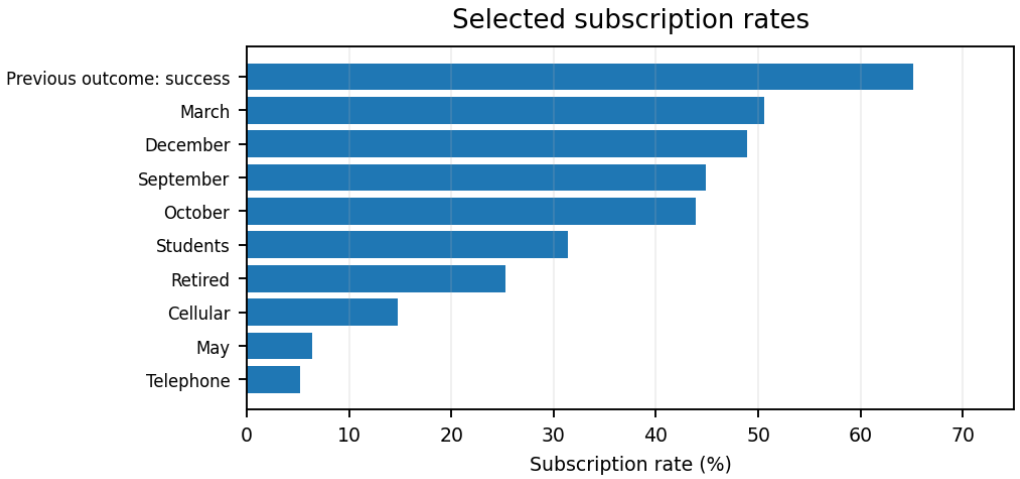
2. Data quality and preparation

The notebook reviewed data types, missing values, duplicate records, unknown categorical values and special values. Potential outliers were flagged using the IQR method and assessed in context rather than removed automatically.

Special code: *pdays = 999* represents customers who were not previously contacted. It should not be interpreted as an ordinary elapsed-day value.

3. Distribution and segment analysis

The analysis examined numerical distributions, skewness, correlations and feature distributions by target class. Campaign, previous-contact count and duration were notably right-skewed in the notebook's summary.



4. Statistical analysis

Chi-square tests: most categorical variables showed statistically significant associations with subscription. Housing and loan were not statistically significant at the 5% level in the notebook's tests.

Mann–Whitney U tests: the notebook reported statistically significant differences between subscription groups for all 10 numerical features. With a large sample, statistical significance should be considered alongside practical effect size.

Cramér's V: previous campaign outcome (*poutcome*) had the strongest categorical association with subscription (approximately 0.3205).

5. Selected subscription rates

Segment / factor	Subscription rate
Previous outcome = success	65.11%
March	50.55%
December	48.90%
September	44.91%
October	43.93%
Students	31.43%
Retired customers	25.26%
Cellular contact	14.74%
May	6.44%
Telephone contact	5.23%

These are descriptive rates from the analysed dataset. They indicate association, not causation. Very small groups (for example, rare default or education categories) should be interpreted cautiously.

6. Business insights and recommendations

Insights. Previous campaign success is strongly associated with subscription; rates vary by month; students and retired customers had relatively high observed rates; and cellular contact outperformed telephone contact in this dataset.

Recommendations. Prioritise follow-up strategies informed by previous campaign outcomes; review seasonal patterns; test targeted approaches for promising customer segments; compare communication channels on both conversion and cost; and monitor results using subscriber counts as well as rates.

7. Conclusion

The exploratory analysis identified useful patterns across customer demographics, campaign history, timing, contact channels and economic indicators. These findings informed the predictive modelling task. This report reflects the analysis recorded in the supplied notebook and does not claim causal effects.